

Functional Requirements Document (FRD)

Medical Appointment Management System (MedSched)

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1. Introduction & Scope

This document specifies the functional requirements for developing a web and mobile software solution designed to streamline and optimize appointment management within a medical practice. The primary objectives of the system are to mitigate appointment no-show rates, accelerate administrative workflows for clinic staff, and provide a frictionless operational experience for healthcare practitioners and patients alike.

2. System Actors & Roles

- **Administrator:** Grants unrestricted system-wide access. Responsible for configuring master data catalogs (practitioners, clinical specialties, branches) and generating global cross-sectional reports.
- **Receptionist / Administrative Staff:** Manages daily schedule operations, patient onboarding, manual booking via walk-ins or phone calls, and real-time waiting room monitoring.
- **Practitioner (Doctor):** Inherits personalized access to their dedicated medical schedule, handles schedule blockages for unexpected absences or symposiums, and reviews patient appointment insights.
- **Patient:** Retains self-service capabilities allowing them to book, cancel, or reschedule appointments asynchronously via the web portal or native mobile applications.

3. Detailed Functional Requirements

3.1. Patient Management Module (FR-01)

Provides a centralized, highly secure repository for keeping records of patient demographic information and basic medical criteria while strictly ensuring data consistency.

- **FR-01.1 Patient Onboarding:** The system must capture essential fields during registration: Full legal name, Date of birth, Biological sex, Primary phone number, Email address, and Known clinical allergies.
- **FR-01.2 Identity De-duplication:** The platform must evaluate uniqueness using a unique government-issued identity registration token or email verification prior to persisting records.

3.2. Agenda & Appointment Control Module (FR-02)

Serves as the core transaction engine of the ecosystem, orchestrating concurrency and calendar logic across healthcare professionals.

Req. ID	Core Function	Detailed Operational Description	Priority
FR-02.1	Slot Booking	Allows selecting a practitioner, clinical specialty, date, and available timeframe. The system shall dynamically lock the chosen slot for up to 5 minutes during checkout to eliminate double-booking.	High
FR-02.2	Availability Validation	Prevents the allocation of scheduling slots outside the practitioner's configured roster shifts or on declared national public holidays.	High
FR-02.3	Cancellation & Rescheduling	Enables patients and front-desk users to cancel or shift appointments up to 24 hours prior to the slot. Vacated slots must immediately reflect as available.	Medium
FR-02.4	Calendar Exception Blocks	Empowers practitioners and administrative managers to explicitly block explicit hourly intervals or complete days due to conferences or leaf of absence.	Medium

3.3. Automated Notifications & Reminders Module (FR-03)

Focuses on event-driven communication to minimize clinic absenteeism through automated loops.

- **FR-03.1 Instant Transactional Alerts:** The platform shall fire an automated email and SMS notification conveying exact appointment parameters immediately following successful confirmation.
- **FR-03.2 Proactive Attendance Verification:** The module must trigger an interactive reminder 24 hours prior to the event, demanding explicit acknowledgement via a single-click action link or structured SMS string response.

3.4. Consultation Flow & Waiting Room Dashboard (FR-04)

Supplies live visualization mechanics to map the real-time physical transit of registered patients within clinic premises.

- **FR-04.1 Appointment Lifecycle Transitions:** Desk operators must be capable of transitioning appointment records across states: *Pending, In Waiting Room, In Consultation, Completed, Cancelled, or No-Show*.
- **FR-04.2 Practitioner Workspace Overview:** The system must supply doctors with an auto-refreshing interface listing incoming individuals sorted by their physical timestamp arrival.

4. Key Non-Functional Requirements

- **Security & Privacy Safeguards (NFR-01):** All patient records must undergo encryption during transmission using robust TLS (HTTPS) protocols, alongside AES-256 standards for structural database encryption at rest, fulfilling rigorous personal health information regulatory standards.

- **System Concurrency (NFR-02):** The infrastructure must reliably handle a baseline threshold of 50 simultaneous clinic administrative accounts while handling up to 500 parallel incoming patient requests without processing delays (target server response latency < 2.0 seconds).